Technical Paper: Companion – A Smart Academic Management Solution for International Students

1. Background Research

1.1 Problem Space Definition

International students in Canada face a unique intersection of academic, cultural, and logistical challenges. Beyond the standard rigors of higher education, I have observed that these students must navigate complex visa regulations, cultural adjustments, and often, the necessity of part-time employment to sustain their living expenses. My “Companion” project addresses a critical challenge: helping students manage their demanding academic schedules and deadlines, especially when they have many other responsibilities.

1.2 Academic and Industry Research Summary

My research consistently indicates that academic deadlines are a primary source of stress for university students, with international students reporting higher levels of anxiety due to the stakes involved (e.g., maintaining GPA for visa compliance) [1]. A study by Howe (2014) highlights that language barriers and sociocultural adjustments further complicate the academic journey for international students in Canada [2].

Furthermore, my analysis of industry trends in educational technology shows a shift toward “edugration”—the integration of education and migration services—yet many existing tools fail to provide the granular, task-based support needed for daily academic survival [3].

1.3 Gaps and Hypotheses

While learning management systems like Moodle hold a lot of important information, I have found they often don’t present it in a way that actively helps students manage their time effectively.

Gap: A lack of integrated, proactive notification systems that account for a student's overall mental workload (including both study and work).

Hypothesis 1: Visualizing deadlines in a unified, high-interruption nudge system will significantly reduce missed assignments.

2. Market Analysis & Target Users

2.1 Competitive Analysis

Solution	Strengths	Weaknesses
Moodle Mobile App	Direct integration, official support.	Complex UI, lacks proactive planning features, visual uniformity causes cognitive overload.
Todoist / TickTick	Excellent task management, cross-platform.	No native LMS integration; requires manual entry of academic deadlines.
Notion	Highly customizable, all-in-one workspace.	Steep learning curve; manual setup for academic tracking is time-consuming.
Companion (My Proposal)	Automated LMS sync, proactive nudges, stress forecasting.	New entry, requires robust API stability.

2.2 Target User Profiles & Personas

Primary User Group: International students enrolled in Canadian post-secondary institutions (e.g., KPU).

User Persona: "Busy Balwinder"

- **Profile:** 22-year-old international student from India, studying Information Technology.
- **Goals:** Maintain a 3.5 GPA, work 20 hours/week at a local cafe, and eventually apply for PGWP.
- **Pain Points:** Forgets deadlines during busy work shifts; feels overwhelmed by the "wall of text" in Moodle dashboards.

2.3 Value Proposition

I aim for Companion to be a helpful academic assistant for international students. By automatically pulling deadlines from Moodle and combining them with a student's personal schedule, I will turn a simple list of dates into a clear, prioritized plan for staying on track.

References

[1] Capelle, J. D. (2022). Deadlines make you productive, but what do they do to your stress? Journal of Academic Psychology. [https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2022.956000/full]

[2] Howe, E. R. (2014). Experiences of International Students at a Canadian University. Education Research International. [https://files.eric.ed.gov/fulltext/EJ1038901.pdf]

[3] University Affairs. (2023). International students as problems and solutions. [https://universityaffairs.ca/opinion/international-students-as-problems-and-solutions/]

3. Technical Feasibility Analysis

3.1 Required Technologies & Frameworks

- **Frontend (MVP):** I will build a web-based application using React for a dynamic and responsive user interface.
- **Backend (MVP):** I will use Supabase for backend services, including real-time database, authentication, and serverless functions.
- **Database (MVP):** I will utilize Supabase PostgreSQL as the primary database for storing user preferences, academic data, and other application-related information.
- **Authentication (MVP):** I will implement Supabase Auth to handle user authentication, providing secure sign-up, sign-in, and session management.

3.2 Data Sources & Integrations

- Manual input, Moodle iCal link, Google/Outlook calendar
- Twilio for calling number
- ElevenLabs or Grok for TTS services
- Gemini for AI support & creating kanban board from imported calendars

3.3 Technical Risks & Constraints

- **Moodle iCal Export Limitations:** Since I am relying on Moodle's iCal export, real-time, two-way syncing with Moodle's internal assignment system won't be possible. This means Companion will primarily be a 'read-only' view of Moodle deadlines, and users won't be able to change Moodle assignments directly through Companion.
- **Data Privacy:** I will need to be very careful with academic data, ensuring I follow privacy rules like PIPEDA in Canada. This means being transparent about what data I use and how it's protected.
- **Build Complexity (MVP):** Keeping Moodle events (from iCal), Google Calendar, and any manually added tasks in sync will be a key challenge for me. I will start with a simpler approach, perhaps prioritizing Google Calendar as the 'source of truth' and focusing on clear display rather than complex two-way syncs for the MVP.

4. Requirements Document

4.1 Problem Statement

I have identified that international students often find it hard to keep up with academic deadlines while also managing their jobs and personal lives. This is because important information is scattered across different platforms, leading to stress and sometimes missed assignments.

4.2 User Requirements

- **Students:** Need to see all their deadlines in one clear place, get timely reminders, and easily add their work hours.

4.3 Functional Requirements

1. **Moodle Calendar Import:** I will allow students to easily import their academic deadlines and events from Moodle using its iCal export feature.
2. **Google Calendar Sync:** I will connect with Google Calendar to show both Moodle-imported events and any tasks students add themselves.
3. **Smart Notifications (MVP):** I will provide simple, customizable reminders for upcoming deadlines, leveraging Google Calendar's notification features.

4. **Dashboard (MVP):** I will create a clear overview that highlights upcoming deadlines and busy periods, helping students anticipate stressful weeks.

4.4 Non-Functional Requirements

- **Performance (MVP):** I aim for calendar data to update quickly enough to be useful, targeting updates within a reasonable timeframe (e.g., under 10-20 seconds for key information).
- **Accessibility:** I will ensure the app is easy to use for everyone, following standard accessibility guidelines.
- **Security:** I will keep all student data safe and private. I will use secure methods like Supabase Auth to avoid storing sensitive credentials.

4.5 Constraints & Assumptions

- **Assumption:** I assume students can access and use Moodle's iCal export feature to get their calendar data.
- **Constraint (MVP):** For the first version, I will mostly use Google Calendar for managing all events. Moodle integration will be a one-way street, importing data through iCal. In later versions, I could explore more direct Moodle connections and support for other learning platforms like Canvas or Blackboard.

5. Project Planning & Gantt Chart

5.1 Semester-Long Plan

I will tackle this project in a few main stages: Research, Design, Building the App, and Documentation. My main goals for the first version include having a basic design ready by late February and getting the core Google Calendar integration working by late March.

Phase	Start Date	End Date	Key Deliverable
Research & Requirements	Jan 10, 2025	Feb 05, 2025	Finalized Requirements Doc
UI/UX Design	Feb 06, 2025	Feb 25, 2025	High-Fidelity Prototypes
Backend Development	Feb 26, 2025	Mar 20, 2025	Moodle API Integration
Frontend Development	Mar 21, 2025	Apr 15, 2025	Functional Mobile App
Integration & Testing	Apr 16, 2025	May 10, 2025	Beta Release & Bug Fixes
Final Documentation	May 11, 2025	May 25, 2025	Final Capstone Report

(Note: See attached gantt_chart.png for a visual representation of the timeline and dependencies.)

6. JIRA Setup

6.1 Board Structure

I will use a simple project management approach, possibly with a tool like Trello or a basic JIRA setup, to keep track of my tasks. I will break the work down into:

- **Epics:** Big goals like “Getting Moodle data into Google Calendar,” “Designing the User Interface,” and “Setting up Reminders.”
- **Features:** Smaller, specific parts of the app, such as “Import Moodle iCal Link” or “Display Upcoming Deadlines.”
- **Tasks & Subtasks:** Detailed steps, like “Set up Supabase Backend” or “Create a simple list view for deadlines.”

6.2 Sprint Structure

I will work in short cycles, perhaps every two weeks, to build and review parts of the app. Each cycle will start with planning what to work on and end with checking if tasks are completed as planned.

7. Subject Matter Experts (SMEs)

I have identified some helpful experts who can guide me on the technical side, understanding student well-being, and how universities operate.

Name	Role/Organization	Relevance	Contribution Method
Kuvan Parhar	Student Counsellor (KPU Alumni)	Expertise in student stress and support systems.	Guidance on stress-forecasting logic and mental health nudges (LinkedIn / Alumni Network).
Shubham Kaniya	Automation Developer	Specialist in workflow automation and notification triggers and reminders.	Technical advice on Moodle API and reminders (Professional Network / Email).
Dr. Jane Doe	Professor of EdTech	Academic research background in LMS integration.	Validation of the foundation and pedagogical impact (University Directory / Office Hours).

8. Mentor Identification & Confirmation

8.1 External Mentor

- **Name:** Shubham Kaniya
- **Role:** Automation Developer
- **Relevance:** Shubham is great at building automated reminder systems and connecting different software using APIs. His technical know-how is a perfect match for what I am trying to achieve with Companion.

8.2 Proof of Confirmation

I am still awaiting his confirmation.

9. AI Acknowledgment

I used ChatGPT as a tool for idea brainstorming and to help frame the content in professional English. It is important to note that all core ideas, written concepts, and the overall direction of this project are my own; the AI was utilized under my direct control to ensure the final text was properly structured and professionally articulated.